

Network Traffic Generator Using eBPF and WebAssembly



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The Current Landscape of Network Testing Tools

Standard tools often force a trade-off between ease of use and capabilities.

iperf / iperf3

 Can measure TCP/UDP throughput.

 Lacks handling of L2/L3 packets.

Cisco TRex (DPDK based)

 Can generate diverse traffic starting from raw L2 frames.

 Barrier to Entry: Requires complex DPDK setup, significant OS configuration changes, specific NIC hardware, and Python dependencies.

The eBPF Approach & Challenge

- eBPF XDP (eXpress Data Path) Live Frames Mode (BPF_PROG_TEST_RUN)
 - 👩 A kernel feature allowing ultra-high-speed packet transmission loop entirely within the kernel space.
- The Challenge
 - 👩 Native Live Frames mode only repeats a single, static packet image.
 - This is unrealistic for real-world testing where traffic needs variation (e.g., different source IPs, port numbers, protocols and length).

Our Solution: xdperf

<https://github.com/takehaya/xdperf>



- WebAssembly (Wasm) plugins

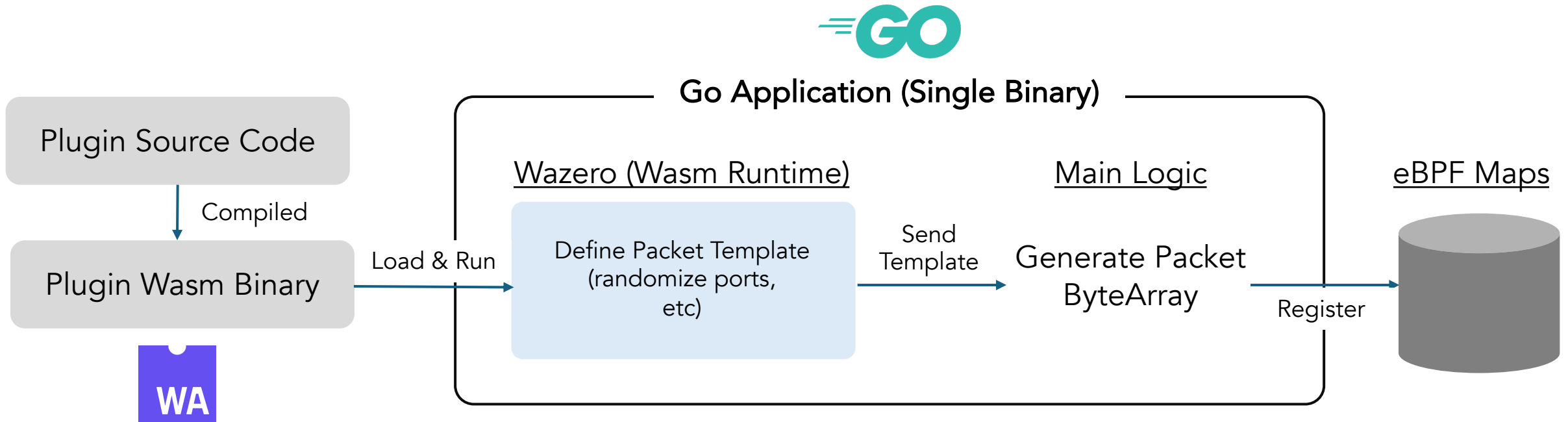


- Test scenarios and packet definitions are written
- Developers can use any language that compiles to Wasm (Rust, TinyGo, etc.).

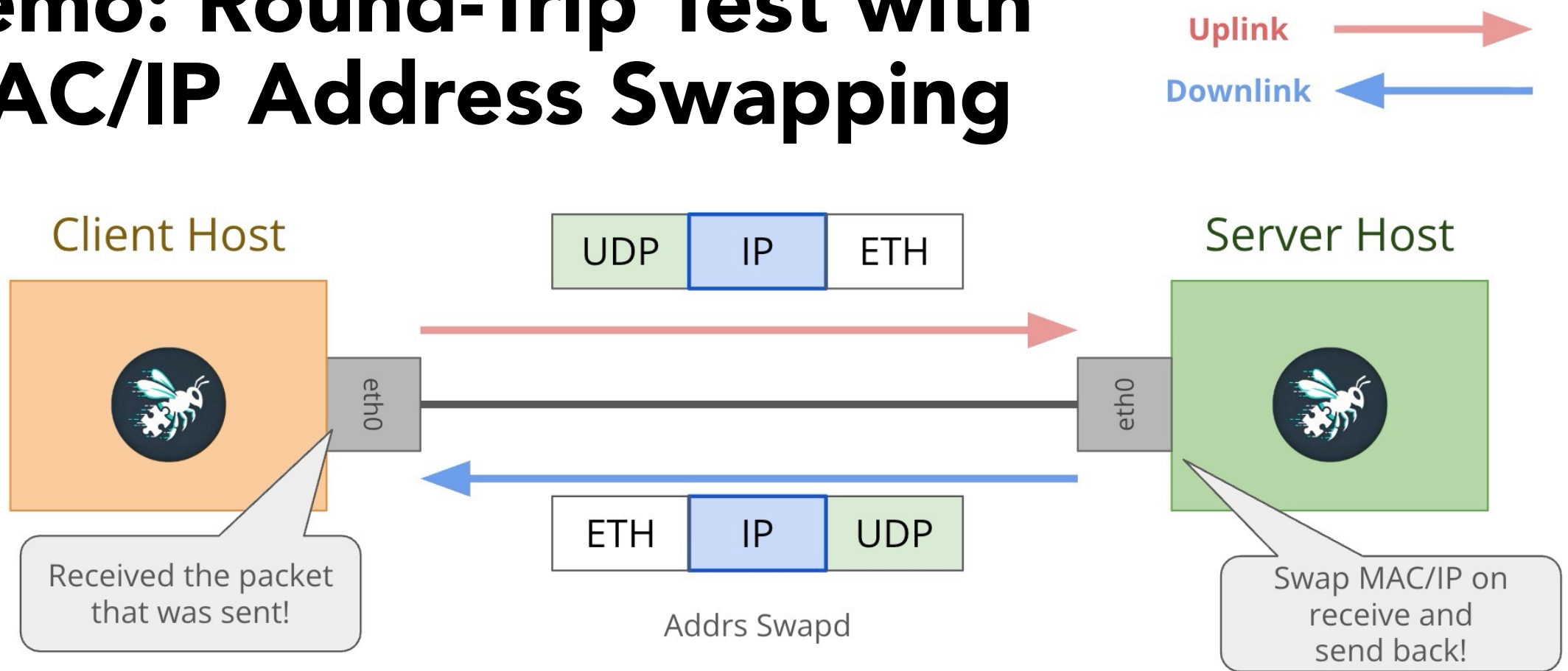
- Go Application

- A single Go application embeds the Wasm runtime (wazero) and manages eBPF interactions.
- Generate packet byte sequences according to the templates passed from Wasm, and register them into an eBPF map.
- Actual packet transmission is handled by XDP in the kernel.

Our Solution: xdperf



Demo: Round-Trip Test with MAC/IP Address Swapping



```
sudo xdperf --recv --plugin  
simpleudp.go --device eth0 --pps  
100 --duration 5s --parallelism 2  
--cfg '{"dst_ip": "10.10.10.11"}'
```

```
sudo xdperf --device eth0  
--send=false --recv=true  
-swap
```

Key Advantages

- Simple and Flexible Setup
 - Does not require Python environments, complex DPDK setups, or invasive OS configurations.
 - Test scenarios can be written in the user's preferred language.
 - No rebuild of the main Go application is required after configuration changes.
- Real World Testing
 - Unlike standard Live Frames, it can generate varied traffic.